

KOSIK OTTÓ

IT Engineering Student — Backend, DevOps/Cloud & Full-stack Development

Budapest, Hungary | +36 50 116 5350 | ifj.kosik.otto@gmail.com | github.com/kosikotto | kosikotto.ddns.net/ko | linkedin.com/in/kosikotto

PROFILE

24-year-old IT engineering student at Óbuda University, specializing in cloud technologies and IT security. Hands-on experience in backend development (C#/.NET, Java), frontend technologies (React, TypeScript, HTML/CSS/JavaScript), as well as server and system administration on both Windows Server and Linux. Passionate about designing and building modern, scalable web applications and the infrastructure behind them — with several independently designed, layered backend systems and a self-managed on-premise server to my name. Also an active team player: since 2024 I have been working as a full-stack developer for O.U.R Team (Óbuda University Racing Team), and I teach programming as a teaching assistant.

SKILLS

Backend & Architecture: C# (ASP.NET Core, Entity Framework, Dependency Injection, OpenAPI)

Java (*Intermediate*): Spring Boot, Project Lombok, OpenAPI, Dependency Injection, REST API, Socket programming

HTML / CSS / JavaScript (*Intermediate*): responsive, cross-browser-compatible frontend development

React (*Intermediate*): component-based frontend development

TypeScript (*Beginner*): type-safe development

Tailwind CSS (*Intermediate*): utility-first styling

Bootstrap (*Intermediate*): responsive UI framework

WordPress (*Beginner*): content management system, custom themes

MySQL (*Intermediate*): relational database design and management

Oracle SQL (*Novice*): queries, basic database management

Databases (other, project experience): PostgreSQL (JSONB), Redis, MinIO, MSSQL, SQLite

Docker (*Beginner*): containerization, Docker Compose

AWS (*Beginner*): Lambda, S3, DynamoDB, API Gateway

Linux (*Intermediate*): Apache2, Samba, self-hosting (Plex, MySQL, WordPress), minor sysadmin tasks

Windows Server (*Intermediate*): Active Directory, file sharing, permission management, quotas, IIS, automated deployment scripts

Git (*Intermediate*): version control

GitHub (*Intermediate*): repository and project management

Version control (other): Gitea

EXPERIENCE

Operations Intern — National Tax and Customs Administration (NAV), IT Department

Feb 2026 – present

- Alongside daily operations tasks, independently designed and built an automation application to replace a manual workflow, on my own initiative.
- PDF data extraction & Excel automation tool (Java, Apache POI, Spire.PDF): extracts structured data from internal PDF documents and dynamically populates formatted Excel templates.

Full-stack Developer — O.U.R Team (Óbuda University Racing Team)

Oct 2024 – present

- Backend- and DevOps-focused work: server-side and operations tasks complemented by frontend development; independently took over and completed abandoned projects.
- Designed and developed serverless APIs in an AWS environment (Lambda, API Gateway).
- Applied Docker-based containerization to give projects reliable, isolated runtime environments.

Teaching Assistant — Problem-Solving Using Programming — Óbuda University

Sep 2025 – Feb 2026

- Taught programming and algorithmic fundamentals to fellow IT engineering students in English, reviewed code and shared problem-solving strategies.

Junior Teaching Assistant — Problem-Solving with Programming — Óbuda University

Sep 2024 – Feb 2025

- Assisted instructors during Hungarian-language lab sessions, reviewed students' code, and occasionally led sessions independently.

PROJECTS

NIDUS – Parts Version Control & Dependency Management System (in progress) — O.U.R Team

Technologies: ASP.NET Core Web API, PostgreSQL (JSONB), MinIO, Docker

- Modular-monolith-based engineering/logistics backend for the Formula Student team; semantic versioning, multi-step approval workflow, BOM aggregation.

Central Authentication & Distribution Portal (in progress) — O.U.R Team

Technologies: Keycloak SSO, RBAC

- Central web hub providing access to internal applications and microservices; role-based access control to separate data across engineering departments.

Custom Forum (in progress) — O.U.R Team

Technologies: C#, Docker

- Backend service for a blog-style internal forum; automated content pipeline that dynamically updates the official website based on published articles and media.

Scalable E-commerce Backend API

Technologies: Java, Keycloak, Redis, OpenAPI (Swagger), Lombok, Docker

- Containerized REST API for a planned webshop; Keycloak-based RBAC, Redis caching for frequently requested data, strict endpoint validation, and OpenAPI documentation.

https://github.com/kosikotto/scalable_backend_api

Self-hosted Portfolio & Reference Server

Technologies: Linux, Docker, PostgreSQL, Redis, MinIO

- Own on-premise server running reference projects in Docker containers; Redis cache and MinIO object storage integrated for backend optimization.

On-premise Infrastructure & Self-hosting

Technologies: Linux, Nginx/Apache, SSL/TLS, Keycloak, DNS/networking

- Full management of a physical server and network: mandatory HTTPS redirection, automated SSL/TLS renewal, custom reverse proxy for subdomains, Keycloak integration to separate public and private services.

Energy Cost Calculator

Technologies: Docker, Docker Compose, Apache2, Vanilla HTML/JS

- Full-stack application calculating energy costs from yearly/monthly consumption data with a custom discount system; frontend and backend in separate containers, reverse proxy + SSL on a self-hosted server.

<https://github.com/kosikotto/Villanyzamlakoltsegek>

Formula Student Inventory Management System / Part-Number Generator

Technologies: ASP.NET Core (MVC), MySQL, Redis, Docker, JavaScript

- System for version-tracking and cataloguing parts for the university's Formula Student team; Redis caching layer to reduce query load.

<https://github.com/kosikotto/raktarkezelo>

Console RPG Game (Shakes & Fidget-style)

Technologies: C#, Entity Framework (DB-First), MSSQL

- Layered architecture (Domain, Application, Infrastructure, UI) with Dependency Injection for scalability and clean code.

<https://github.com/kosikotto/rpg-console-game>

PDF Data Extraction & Excel Automation Tool

Technologies: Java, Apache POI, Spire.PDF

- Custom automation solution: extracts structured data from PDF documents and dynamically populates Excel templates. (Built on request for my employer; source code is not public.)

React Movie Catalogue

Technologies: React, TypeScript, Tailwind CSS

- Learning project to master the React / TypeScript ecosystem, with a well-structured, responsive UI.

<https://github.com/kosikotto/react-filmkatalogus>

Cross-platform Weather App

Technologies: .NET MAUI, MVVM, SQLite, OpenWeather API

- Mobile-optimized weather app built with the MVVM pattern; real-time and 5-day forecasts, GPS-based location detection, offline caching via SQLite.

https://github.com/kosikotto/T0Y9UZ_Kosik_Otto_Feleves

Interactive Solar System Model (WIP)

Technologies: HTML5, CSS3, Vanilla JavaScript

- Framework-free, interactive planetary system visualization; practice with complex animations and relative sizing (vw/vh).

<https://github.com/kosikotto/solar-system>

Basic Responsive Website

Technologies: HTML5, CSS3, Vanilla JavaScript

- Fully responsive website built without external libraries; practice with DOM manipulation and media query fundamentals.

<https://github.com/kosikotto/kosikotto-cv>

First Web Development Project

Technologies: HTML5, CSS3, Vanilla JavaScript

- Responsive, animated UI with a preloader and a scroll-reactive navigation bar — my first project going beyond a simple 'Hello World'.

<https://github.com/kosikotto/sanyibrigad>

EDUCATION

BSc in Engineering Informatics — Cloud Technologies & IT Security specialization — Óbuda University, John von Neumann Faculty of Informatics

Sep 2023 – present

IT Systems Operator (vocational qualification) — Budapest Technical SZC Egressy Gábor Secondary School

Sep 2022 – Jun 2023

Bilingual technical secondary school — Budapest Technical SZC Egressy Gábor Secondary School

Sep 2017 – Jun 2022

CERTIFICATIONS & LANGUAGES

- 3142/9 FEOR — Computer System Maintainer
- 54-481-06 — IT Systems Operator
- English — C1 (advanced, complex exam)
- Hungarian — native